XI(SEAN) CHEN

xchen4@umass.edu ◇ Homepage ◇ Linkedin

EDUCATION

University of Massachusetts Amherst — **Amherst, MA**
- PhD in Computer Science (GPA: 3.9/4) — 2019-2025

Central South University — **Changsha, China**
- B.E. in Computer Science (GPA: 3.9/4) — 2014-2018

EXPERIENCE

Pretraining & Scaling – Amazon AGI Foundations — **Sunnyvale, CA**
Applied Scientist — 2025.7 - present
- Optimized pre- and mid-training recipes by scaling law for Amazon's base models (**hundreds B, 10k+ gpus**).
- Diversified reasoning path by contrastive learning and self-supervised token-efficient curriculum for mid-training.
- Unified datamix within a framework that stabilizes adaptive attention and distribution-constrained optimization.
- Refined evaluation pipelines across foundation models with shot/prompt/inference sensitivity analysis.

Amazon AGI Foundations — **Seattle, WA**
Applied Scientist Intern — 2024.7 - 2024.10
- Applied parameter efficient LLM training with Quantized LoRA for scalable model adaptation.
- Devised a model merging framework by distilling activation to accelerate multi-task LLM learning schedule.
- Implemented adaptive online learning with sample re-weighting for continuous LLM improvement.

University of Massachusetts Amherst — **Amherst, MA**
Research Assistant — 2019.9 - 2025.6
- Developed **the largest multilingual news similarity dataset**, using active learning for retrieval and ranking.
- Built a SOTA transformer to infer global news graph with trillions of edges for event detection, recommenedation.
- Designed an efficient cluster-aware LLM judge framework for online event extractions from temporal graph.

Z.AI — **Beijing, China**
Research Scientist (Directly reported to Jie Tang) — 2017.6 - 2019.4
- Modeled the dynamics of opinions and personalities with probabilistic graph time series.
- Devised an extended EM algorithm for learning the graph time series and get competitive performance.
- Searched and rank scholars with heterogeneous random forest (deployed on prestigious AI platform Aminer).
- Clustered and visualize temporal scientific topics with refined genetic algorithm.

PROJECT

Nova Foundation Model Training Optimization (**Artificial Analysis Intelligence Index: 62**)
- Optimized pre&mid-training recipes for Amazon's Nova base models, supporting official production release.
- Designed efficient TB-scale data preparation pipelines to support high-throughput generation.
- Analyzed training dynamics and failure modes through ablations of hyperparamters and data mixtures.

Reasoning Path Diversification in Mid-Trainin (**+16% coding performance**)
- Rewrote noisy webcrawl data with LLM judge to summarize reasoning information.
- Designed contrastive knowledge distillation strategies to generate diversified reasoning data.
- Quantified data reasoning depth with Essential-web classifier to optimize mid-training recipe.

Merge LLM with Online Activation Alignment (**+10% math performance**)
- Parameter-efficient fine-tuning with data mixing and target evaluation for domain expert models.
- Developed an activation alignment framework to merge LLMs across diverse tasks (math, coding, medical).
- Adapted activation weight for LLM online learning during parameter-efficient training and sampling.

Cross-Model Evaluation and Benchmarking Framework (**Aligned up to 17% evaluation gap**)
- Standardized evaluations across frontier foundation models with shot/prompt robustness check.
- Enabled rigorous cross-model comparison through unified metrics and analysis workflows.

- Generated evaluation insights to inform model selection and training roadmap decisions.

Efficient Cluster-wise LLM Judge for Event Identifications from Poll (**10x data efficiency**)

- Designed a cluster-aware active learning framework to detect temporal events from X poll texts.
- Modeled the evolution of social discourse by graph-based topic tracking and statistical analysis.

Global Multilingual News Graph Learning with Transformers

- Devised a global news retrieval system, generating trillion-scale relations with dense embeddings and indexing.
- Built rank models for similar multilingual news using rules-based and active learning classifiers.

SELECTED PUBLICATION

Amazon Nova 2: Multimodal reasoning and generation models
Amazon AGI Foundations. Technique Report. 2025.

Adaptive Activation Alignment for Model Merging
Xi Chen, Akshaya Shanbhogue, Yue Zhang. To appear.

Identifying and Investigating Global News Coverage of Critical Events Such as Disasters and Terrorist Attacks
Xi Chen*, Erica Cai*, Reagan Keeney, Ethan Zuckerman, Brendan O'connor, Przemyslaw Grabowicz. International AAAI Conference on Web and Social Media (**ICWSM**), 2025.

Efficient Cluster-aware Large Language Model Learning for Event Identification from Social Polls
Xi Chen, Mayank Bumb, Vishal Kalakonnavar, Przemyslaw Grabowicz. To appear, 2025.

Global News Synchrony and Diversity During the Start of the COVID-19 Pandemic
Xi Chen, Scott Hale, David Jurgens, Mattia Samory, Ethan Zuckerman, Przemyslaw Grabowicz. International World Wide Web Conference (**The Webconf**), 2024.

A Multilingual Similarity Dataset for News Article Frame
Xi Chen, Scott Hale, David Jurgens, Mattia Samory, Przemyslaw Grabowicz. International AAAI Conference on Web and Social Media (**ICWSM**), 2023.

Multilingual News Article Similarity
Xi Chen, Ali Zeynali, Chico Camargo, Fabian Flock, Devin Gaffney, Przemyslaw Grabowicz, Scott Hale, David Jurgens, Mattia Samory. International Workshop on Semantic Evaluation @ **NAACL**, 2022.

SKILL

Pre&mid-training: training dynamics, scaling law, parameter scaling, datamix optimization, synthetic data.
Post-training: Multi-teacher on-policy knowledge distillation, Parameter-efficient fine-tuning, Model merging.
Search & Retrieval: Retrieval-augmented generation(RAG), similarity search, retrieval models, Semantic indexing.
Recommendation Systems: Graph modeling, Active learning for ranking, Multilingual content recommendations.
Machine Learning & Optimization: Event detection, Time-series analysis, Probabilistic modeling.

SERVICE

Invited Talk:

- Modeling Global Events with LLMs, Pittsburgh, 2026. Host: Ruslan Salakhutdinov & Yaser Sheikh (CMU)
- Multilingual News Article Similarity, Chicago, 2022. Host: Chenghao Tan (UChicago)

Program Member:

- NAACL(2023)
- ICWSM(2023, 2024, 2025, 2026)
- SEMEVAL (2022, 2023, 2024)
- IC2S2 (2022, 2023, 2024, 2025)

HONOR

NSF Student Travel Award, 2024
UMass CICS best portfolio finalist, 2023.
Nomination for Chinese Exceptional Student (1/6500 ISchool undergrads and grads), 2018.
Ranked global Top50 of Autochess players (50/millions, a world-class strategy game for playing DOTA on chesstable).